

CCNA 200-301 V2.0

05 IPv6 Addressing

Part I — Foundations

EXAM TOPICS

1.4

LECTURER

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What you will be able to do

- **Abbreviate** and expand an IPv6 address correctly, applying both compression rules without ambiguity.
- **Identify** the address types the exam names: global unicast, link local, unique local, multicast, and the solicited-node address.
- **Construct** an interface identifier by modified EUI-64, which the blueprint calls out explicitly.
- **Size** a prefix, and **diagnose** the assignment faults that stop an IPv6 host communicating.

Before we start

Chapter 4 — the ideas of network part, host part and prefix length carry over unchanged.
Chapter 3 for MAC addresses, which EUI-64 is built from.

Vocabulary for this chapter

global unicast

link local

unique local

multicast

solicited-node

anycast

modified EUI-64

SLAAC

router advertisement

neighbour discovery

dual stack

prefix

Each is defined where it first matters — not here.

What v2.0 asks, and what it dropped

This is what the blueprint actually asks for

Topic **1.4** says *troubleshoot* IPv6 address configuration, assignment and prefix sizing, and narrows the scope to “unicast and modified EUI 64”. The previous blueprint asked you to compare every address type in detail; this one does not. Concentrate on getting an address onto an interface, understanding where the link-local one came from, and spotting a wrong prefix.

The two compression rules

- **Drop leading zeros** in any group. `0db8` becomes `db8`; `0000` becomes `0`. Trailing zeros are never dropped.
- **Replace one run of all-zero groups with ::**. You may do this **once only**, because a second `::` would make the address ambiguous — there would be no way to know how many zeros each stood for.

Compressing an address

Start with the full form:

2001:0db8:0000:0000:0000:ff00:0042:8329

- Drop leading zeros in each group: 2001:db8:0:0:0:ff00:42:8329
- Replace the longest run of zero groups with :: 2001:db8::ff00:42:8329

Expanding is the same in reverse. Given 2001:db8::1, count the groups present — two before the :: and one after, so three — and the :: must therefore stand for five zero groups:

2001:0db8:0000:0000:0000:0000:0000:0001.

There is no broadcast in IPv6

IPv6 has no broadcast address at all. Where IPv4 would broadcast, IPv6 uses a multicast group — usually `ff02::1`, all nodes. This is why ARP does not exist in IPv6: its job is done by neighbour discovery using the solicited-node multicast address, which reaches only the handful of hosts whose addresses end the same way rather than every device on the segment.

Turning a 48-bit MAC into a 64-bit interface ID

- Split the MAC in half.
- Insert `fffe` in the middle.
- **Flip the seventh bit** of the first byte — the universal/local bit. In hex this means adding or subtracting 2 from the second hex digit.

EUI-64 from 00d0.58a1.0201

- Split: 00d0.58 and a1.0201.
- Insert fffe: 00d0:58ff:fea1:0201.
- Flip the seventh bit of 00. In binary 00 is 0000 0000; flipping the seventh bit gives 0000 0010, which is 02.
- Interface ID is 02d0:58ff:fea1:0201. With a prefix of 2001:db8:acad:10::/64 the full address is 2001:db8:acad:10:2d0:58ff:fea1:201.

The quick check: an address containing ff:fe in the middle of its host portion was almost certainly built by EUI-64.

Ipv6 on a router interface, three ways

Configure

```
! Routing must be enabled globally first -- this is the step most often  
! forgotten, and without it the router will not forward or advertise.  
ipv6 unicast-routing  
!  
interface GigabitEthernet0/0  
  description User VLAN 10 gateway  
  ! 1. A fully specified address.  
  ipv6 address 2001:db8:acad:10::1/64  
  ! 2. Or let the device build the host part by EUI-64.  
  ipv6 address 2001:db8:acad:20::/64 eui-64  
  ! 3. A link-local address is created automatically, but you can pin it  
  ! to something memorable -- worth doing, because routing protocols  
  ! show it as the next hop.  
  ipv6 address fe80::1 link-local  
  no shutdown
```

show ipv6 interface brief

```
R1# show ipv6 interface brief
```

```
GigabitEthernet0/0    [up/up]  
    FE80::1  
    2001:DB8:ACAD:10::1  
    2001:DB8:ACAD:20:2D0:58FF:FEA1:201  
GigabitEthernet0/1    [administratively down/down]  
    unassigned
```

How a host gets an address: SLAAC versus DHCPv6

A host sends a router solicitation to `ff02::2`; a router replies with a **router advertisement** carrying the prefix. Then either:

- **SLAAC** — the host builds its own address from the advertised prefix plus an interface ID it generates. No server involved.
- **DHCPv6** — the advertisement tells the host to ask a server, which supplies the address and other options.

A host with only a `fe80::` address and no global one has heard no router advertisement. That is the IPv6 equivalent of an APIPA address, and it points at the router, not the host.

Configuring IPv6

- **Omitting `ipv6 unicast-routing`.** Interfaces come up and addresses look right, but the router neither forwards nor advertises.
- **Using `::` twice.** It is not merely discouraged, it is invalid, because the address can no longer be expanded uniquely.
- **Dropping trailing zeros.** `2001:db8:1000::` cannot be shortened to `2001:db8:1::`. Only *leading* zeros go.
- **Assuming a prefix other than /64 for a LAN.** SLAAC requires a 64-bit interface ID, so a LAN prefix longer than /64 breaks autoconfiguration. Point-to-point links are the usual exception.

Dual-stack host reaches IPv4 destinations only.

A workstation has a working IPv4 address and, for IPv6, only `fe80::2d0:58ff:fea1:201`. It cannot reach any IPv6 destination, on or off the segment.

Dual-stack host reaches IPv4 destinations only.

What would you check first — and what do you expect to see?

Why it is doing that

Diagnosis. A link-local address is generated by the host itself and proves nothing except that the interface is up. The absence of a global address means no router advertisement is being received. Either no router is configured for IPv6 on that VLAN, or `ipv6 unicast-routing` is missing, or the router interface has no global prefix to advertise.

And the lesson

Fix. On the gateway, confirm `ipv6 unicast-routing` is present in the running configuration, then check `show ipv6 interface GigabitEthernet0/0` shows a global address and that router advertisements are not suppressed. Bring the prefix back and the host will build its own address within seconds — no change is needed on the host at all.

Dual-stack a small network

Topology. One router, one switch, two PCs, addressed for both protocols.

1. Enable `ipv6 unicast-routing` and give `Gi0/0` the address `2001:db8:acad:10::1/64` and link-local `fe80::1`.
2. Set both PCs to obtain IPv6 automatically. Record the addresses they build and confirm they contain `ff:fe`.
3. Compute by hand what each PC's EUI-64 address should be from its MAC, and check your answer against the device.
4. Ping between the PCs using IPv6, then ping the router's link-local address from a PC. Note that the link-local ping needs no global addressing at all.

Dual-stack a small network (continued)

1. Remove `ipv6 unicast-routing` and observe what happens to the PCs after their addresses expire.
2. **Extension.** Configure a second VLAN with prefix `2001:db8:acad:20::/64` and route between the two.

CHECKPOINT 1 of 3

Compress 2001:0db8:0000:0012:0000:0000:0000:00ab fully.

Discuss · then advance

Checkpoint 1 of 3

Compress 2001:0db8:0000:0012:0000:0000:0000:00ab fully.

2001:db8:0:12::ab. Leading zeros drop in every group; the single run of three all-zero groups becomes ::. Note the fourth group is 0, not part of the :: — only one run may be compressed, and the longer one wins.

CHECKPOINT 2 of 3

A host shows only an fe80:: address. What is missing, and on which device would you look?

Discuss · then advance

Checkpoint 2 of 3

A host shows only an fe80:: address. What is missing, and on which device would you look?

A global unicast address, which means no router advertisement is being received. Look at the router or layer 3 switch for the segment: is `ipv6 unicast-routing` enabled, and does the interface have a global address to advertise a prefix from?

CHECKPOINT 3 of 3

Build the modified EUI-64 interface identifier for MAC
aabb.cc00.0100.

Discuss · then advance

Checkpoint 3 of 3

Build the modified EUI-64 interface identifier for MAC `aabb.cc00.0100`.

Split the MAC, insert `fffe`, and flip the seventh bit: `aa` becomes `a8`. The result is `a8bb:ccff:fe00:0100`.

What to take away

- Drop leading zeros; use `::` once, for the longest zero run.
- `2000::/3` global, `fe80::/10` link local, `fd00::/8` unique local, `ff00::/8` multicast. There is no broadcast.
- EUI-64: split the MAC, insert `fffe`, flip the seventh bit.
- `ipv6 unicast-routing` is a separate, easily forgotten step.
- LANs are /64 because SLAAC needs a 64-bit interface ID.
- Only a link-local address means no router advertisement was heard — investigate the router, not the host.

Where we got to

- Drop leading zeros; use :: once, for the longest zero run.
- 2000::/3 global, fe80::/10 link local, fd00::/8 unique local, ff00::/8 multicast. There is no broadcast.
- EUI-64: split the MAC, insert fffe, flip the seventh bit.
- ipv6 unicast-routing is a separate, easily forgotten step.

NEXT

Chapter 6 — TCP, UDP and the Diagnostic Toolkit